

A Hybrid TimeMixer-XGBoost Framework for Multi-Horizon PM2.5 Forecasting Using Historical PM2.5 Time-Series Data

Thu Le (0009-0008-3480-8311)¹ Nhi Tran (0009-0001-1047-5080)¹ Han Dong
(0009-0002-2532-8099)¹ Ngan Pham (0009-0001-1089-2626)¹

Faculty of Information Technology, FPT University, Ho Chi Minh City, Vietnam

Emails: thulvm@fpt.edu.vn; nhikl12006@gmail.com; donghan010106@gmail.com; tunganphamhoang@gmail.com

Abstract Accurate multi-horizon PM2.5 forecasting is difficult because hourly monitoring records combine persistence, periodicity, abrupt episodes, missing observations, site heterogeneity, and measurement-regime changes. This study evaluates a calibration-gated TimeMixer-XGBoost framework at five UK-Air stations and horizons of 1, 6, 12, 24, and 48 hours. The residual stage removes calibration-period median bias, fits horizon-specific XGBoost models with pseudo-Huber loss and bounded corrections, selects a correction scale on a later held-out calibration subset, and applies rolling calibration using only errors observable at each forecast origin. The benchmark includes persistence, seasonal-naïve, direct XGBoost, DLinear, LSTM, TimeMixer, ungated robust correction, gated static correction, and gated rolling correction. Neural experiments use seeds 0-4 and a 30-epoch maximum under separate training, early-stopping, residual-fit, gate-selection, and locked-test eras. The confirmatory experiment completed five stations, five seeds, and a 30-epoch maximum. Relative to TimeMixer, the gated rolling hybrid reduced pooled MAE at all five horizons, with improvements ranging from 0.84% at 48 hours to 42.58% at 1 hour. Calibration gating suppressed 56.8% of station-seed-horizon corrections, substantially limiting the over-correction observed in the ungated ablation. Nevertheless, the hybrid won 0 of 25 comparisons against the strongest matched non-hybrid comparator; its mean relative MAE improvement under that stricter comparison was -23.33%. LSTM obtained the lowest pooled MAE (3.271). The results support gating as an effective TimeMixer correction safeguard but not a superiority or state-of-the-art claim.

Keywords: PM2.5; TimeMixer; XGBoost; multi-horizon forecasting; air quality; residual learning; time-series forecasting; reproducibility

1 Introduction

Air-quality forecasts must provide useful lead time while remaining reliable during pollution episodes and incomplete monitoring. This matters because particulate exposure is linked to respiratory and cardiovascular harm [1]. Hourly PM2.5 is difficult to forecast because persistence, calendar cycles, abrupt events, missingness, and instrument changes occur in the same record.

TimeMixer represents multiscale temporal structure [2], whereas XGBoost models nonlinear interactions among lagged, rolling, and regime features [3]. We therefore use TimeMixer for the base trajectory and XGBoost for horizon-specific residuals. Median de-biasing, pseudo-Huber loss, bounded corrections, held-out gating, and rolling calibration protect against unstable residual corrections.

Using 168 origin-available hours, the study forecasts 1, 6, 12, 24, and 48 hours without future weather or traffic inputs. Five UK stations, DLinear, LSTM, tree and naïve baselines, seeds 0-4, and a 30-epoch maximum test whether the hybrid improves both TimeMixer and the strongest matched comparator.

2 Related Work

LSTM remains a standard sequence comparator [4], while decomposition-based models show the value of separating temporal modes [5]. PatchTST uses local subsequences [6]; TimeMixer instead downsamples and mixes multiple scales. XGBoost complements these sequence models by targeting structured residual error rather than relearning the full trajectory.

Table 1: Station-specific data limitations identified before model fitting.

Station	Type	Available-history issue	Pre-specified treatment
MY1	Urban traffic	2020 coverage 77.81%; 69-day gap	Preserve long gap; reject affected windows
BIR_A4540	Urban traffic	Record begins in 2021	Report shorter training history
HP1	Urban background	No observations in 2016-2018	Treat as unavailable, not imputed
CHB	Rural background	2024 coverage 73.44%; 561 negatives	Mask negatives; preserve unresolved gaps
KC1	Urban background	Instrument-regime transition	Retain instrument type as context

Because accuracy changes with lead time [7], results are reported by horizon using chronological, common-origin comparisons. MAE is primary [8]; paired inference accounts for serial dependence [9, 10]. All reported values are generated from saved experiment outputs.

3 From Historical PM2.5 Time Series to Multi-Horizon Residual Correction

The pipeline separates raw evidence from modeling values, short-gap interpolation from long-gap preservation, base-model training from residual calibration, and model selection from locked testing.

3.1 Setup

The confirmatory dataset contains five UK-Air stations selected to represent different monitoring environments: London Marylebone Road (MY1, urban traffic), Birmingham A4540 (BIR_A4540, urban traffic), London Honor Oak Park (HP1, urban background), Chilbolton (CHB, rural background), and London North Kensington (KC1, urban background). Each station is processed independently with the same schema, cleaning rules, forecast horizons, chronological boundaries, seeds, and evaluation metrics. Dataset paths and site types are fixed in `data/stations.csv`.

UK-Air exports separate date and hour-ending time fields. During ingestion, the pipeline preserves source filename, source row identity, raw timestamp text, raw concentration text, and status metadata. A source time of 24:00 is normalized to 00:00 of the following day. The target is hourly PM2.5 mass concentration in $\mu\text{g m}^{-3}$.

These differences imply unequal amounts of usable training history in the primary analysis. The primary design uses every station’s legitimately available history because this reflects the operational archive. A pre-specified common-history sensitivity analysis instead restricts all training inputs and targets to 1 January 2021 through 31 December 2022, while leaving the 2023 validation, January-June 2024 calibration, and locked test period unchanged.

3.2 Auditable cleaning and the missingness operator

The raw file is immutable. Cleaning creates three concentration concepts. The *raw* field preserves source evidence. The *observed* field contains numeric, non-negative measurements before imputation. The *clean* field adds only approved short-gap interpolation. Each row carries flags for timestamp validity, missing or non-numeric source values, negative values, duplicate timestamps, inserted hourly-grid rows, interpolation, and preserved long gaps.

The audit identifies 8,087 missing or non-numeric readings before negative handling and 238 additional numeric values below zero. Negative values are masked only in the modeling field; the raw value remains auditable. Duplicate timestamps, if present, are resolved deterministically after sorting by timestamp and source row. The series is then reindexed to an exact one-hour grid so that a missing row and a present row containing “No data” have the same modeling implication while remaining distinguishable in the audit.

Instrument information is parsed from the status field and encoded as TEOM FDMS = 0 and BAM = 1. Instrument metadata may be propagated across grid-only rows because this describes the observation regime,

Table 2: Verified quality profile of the supplied London Marylebone Road record.

Audit item	Value
Raw hourly rows	87 672
Missing/non-numeric readings	8 087
Negative numeric measurements	238
Longest invalid run	1 656 h (69 d)
2020 valid coverage after negative exclusion	77.81%
Instrument transition	22 Dec 2021, 12:00 UTC
Primary short-gap interpolation limit	3 h

whereas concentration values are not propagated. Status metadata identify a TEOM FDMS-to-BAM transition at 12:00 UTC on 22 December 2021.

For each contiguous invalid run, the cleaning stage records start time, end time, duration, class, and the number of inserted rows. Linear interpolation is allowed only when a run is internal, bounded by observed values on both sides, and no longer than three hours. Runs of 4-24 hours, 25-168 hours, and more than seven days remain missing. The maximum invalid run is 1,656 hours, exactly 69 days, and is therefore preserved. Any model sequence or requested target containing unresolved missingness is rejected and counted.

3.3 Origin-safe predictors and a leakage invariant

Calendar predictors contain hour, day of week, month, day of year, weekend status, and sine/cosine encodings for hour, weekday, and annual position. The direct XGBoost baseline uses PM2.5 lags at 1, 2, 3, 6, 12, 24, 48, 72, and 168 hours; rolling mean, standard deviation, minimum, maximum, and median over 3, 6, 12, 24, 48, and 168 hours; one- and 24-hour differences; a 24-hour ratio; cyclic calendar variables; and instrument type.

Every rolling statistic is computed from the series after a one-hour shift. Thus, a rolling feature attached to origin t contains values no later than $t - 1$. The current PM2.5 value y_t remains available as the last element of the 168-hour sequence and to persistence. This convention corresponds to issuing a forecast after the latest observation has arrived.

Proposition 1 (Origin-safety of the constructed predictors). *For a forecast origin t , every sequence, lag, rolling, calendar, instrument, and missingness predictor constructed by the pipeline is measurable from timestamps no later than t . Every target is strictly later than t and is constrained to remain inside its assigned chronological partition.*

Proof. The sequence terminates at the origin; lag features are explicit backward shifts; rolling summaries are computed from a series shifted by one hour; calendar, instrument, and missingness context are read from the origin or its past; and the window constructor accepts targets only at positive configured horizons. A partition-specific target-end constraint rejects any origin whose furthest requested target would cross the partition boundary. Therefore, no target-side value enters a predictor and no target crosses into a later experimental era. $\square$

This invariant is important because multi-horizon sequence windows overlap heavily. A random split would place nearly identical historical contexts on both sides of a boundary, whereas chronological partitions preserve the information set that would exist at deployment.

3.4 Forecasting problem

Let y_t denote hourly PM2.5 and let

$$\mathbf{x}_{t-L+1:t} \in \mathbb{R}^{L \times p} \quad (1)$$

be the origin-available sequence with $L = 168$ hours and $p = 9$ channels. The sequence channels are cleaned PM2.5, a missingness mask, instrument type, and cyclic hour, weekday, and annual encodings. For horizon

Table 3: Core notation.

Symbol	Meaning
y_t	observed hourly PM2.5 at time t
L	input history length, fixed at 168 h
$\mathcal{H}$	direct horizon set $\{1, 6, 12, 24, 48\}$ h
$\mathbf{x}_{t-L+1:t}$	origin-available multichannel sequence
$\hat{\mathbf{y}}_t^{TM}$	TimeMixer base forecast vector
$\mathbf{e}_t$	flattened multiscale TimeMixer embedding
$\mathbf{z}_t$	calendar, instrument, and missingness context
$r_{t,h}$	TimeMixer residual at horizon h
$g_h(\cdot)$	XGBoost residual model for horizon h
γ_h	held-out calibration correction gate for horizon h

set $\mathcal{H} = \{1, 6, 12, 24, 48\}$, the direct task is

$$\hat{\mathbf{y}}_t = (\hat{y}_{t+1}, \hat{y}_{t+6}, \hat{y}_{t+12}, \hat{y}_{t+24}, \hat{y}_{t+48}). \quad (2)$$

Predictions are generated simultaneously and are not recursively fed back. Direct prediction avoids recursive error accumulation and permits one residual correction function per horizon.

3.5 Compact TimeMixer base forecaster

The implementation is a compact research adaptation of the core multiscale ideas in TimeMixer rather than a claim of line-for-line equivalence to the full benchmark implementation [2]. For scales $s \in \{1, 2, 4, 8\}$, the input sequence is average-pooled with stride s to produce $\mathbf{x}^{(s)}$. A centered moving average of kernel 25 estimates a smooth component $\mathbf{t}^{(s)}$, while

$$\mathbf{q}^{(s)} = \mathbf{x}^{(s)} - \mathbf{t}^{(s)} \quad (3)$$

acts as a residual seasonal component. Separate two-layer MLP encoders map $\mathbf{q}^{(s)}$ and $\mathbf{t}^{(s)}$ to 64-dimensional summaries. Their concatenation is fused into a scale representation $\mathbf{e}^{(s)}$.

Two mixer blocks alternate residual MLP transformations across the scale axis and the hidden-feature axis with layer normalization. A linear prediction head at each scale produces five horizon forecasts. Learned softmax weights combine them:

$$\hat{\mathbf{y}}_t^{TM} = \sum_s \alpha_s f_s(\mathbf{e}_t^{(s)}), \quad \alpha_s = \frac{\exp(a_s)}{\sum_j \exp(a_j)}. \quad (4)$$

The mixed representations are flattened into

$$\mathbf{e}_t = [\mathbf{e}_t^{(1)}; \mathbf{e}_t^{(2)}; \mathbf{e}_t^{(4)}; \mathbf{e}_t^{(8)}] \quad (5)$$

for residual calibration. Huber loss, AdamW optimization, gradient clipping, and early stopping are used. Input and target standardizers are fitted on training windows only.

3.6 Horizon-specific residual XGBoost

The residual-calibration interval is divided chronologically into an earlier residual-fit subset $\mathcal{C}_f$ (70%) and a later gate-selection subset $\mathcal{C}_g$ (30%). For each horizon h , the residual is first centred by the median b_h

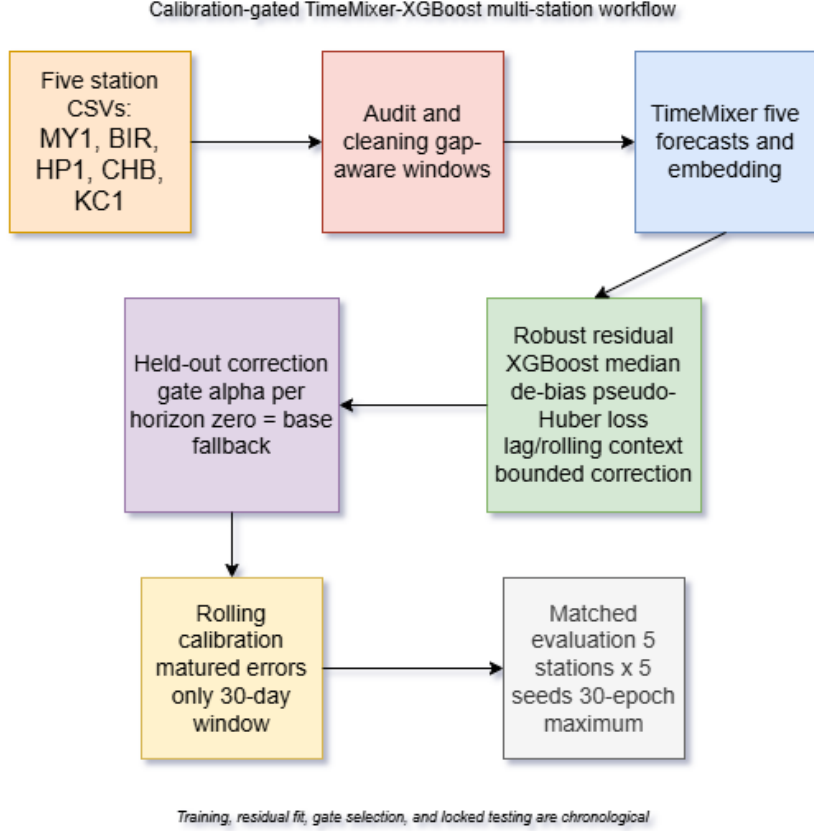


Fig. 1: End-to-end framework. TimeMixer supplies the multi-horizon base forecast; a later calibration interval fits and gates the robust residual correction before locked testing.

estimated only on $\mathcal{C}_f$, after which a separate robust booster learns the remaining error:

$$r_{t,h} = y_{t+h} - \hat{y}_{t,h}^{TM}, \quad b_h = \text{median}_{t \in \mathcal{C}_f}(r_{t,h}), \quad (6)$$

$$\hat{u}_{t,h} = g_h(\hat{\mathbf{y}}_t^{TM}, \mathbf{e}_t, \mathbf{z}_t), \quad (7)$$

$$\hat{c}_{t,h} = b_h + \text{clip}(\hat{u}_{t,h}, q_h^-, q_h^+), \quad (8)$$

$$\gamma_h = \arg \min_{\gamma \in \Gamma} \sum_{t \in \mathcal{C}_g} |y_{t+h} - (\hat{y}_{t,h}^{TM} + \gamma \hat{c}_{t,h})|, \quad (9)$$

$$\hat{y}_{t,h}^{static} = \hat{y}_{t,h}^{TM} + \gamma_h \hat{c}_{t,h}, \quad (10)$$

$$\hat{y}_{t,h}^{HYB} = \hat{y}_{t,h}^{static} + \lambda_t \text{median}(\mathcal{E}_{t,h}). \quad (11)$$

Here $\Gamma = \{0, 0.10, 0.25, 0.50, 0.75, 1\}$. The explicit choice $\gamma_h = 0$ falls back to the TimeMixer base when the correction does not improve held-out calibration MAE. The context vector $\mathbf{z}_t$ contains the origin-safe lag and shifted rolling features used by direct XGBoost, instrument type, cyclic calendar variables, missingness state, gap length, interpolation status, and measurement-quality flags. The complete base forecast vector and latent embedding are also supplied to every horizon model.

Each residual booster uses pseudo-Huber loss, 600 trees, maximum depth 5, learning rate 0.04, row and feature subsampling of 0.8, minimum child weight 3, and L_1/L_2 regularization. Correction bounds (q_h^-, q_h^+) are estimated on $\mathcal{C}_f$. The rolling set $\mathcal{E}_{t,h}$ contains only errors from the previous 30 days whose target timestamps are no later than origin t ; shrinkage λ_t prevents unstable updates when few errors are available. Thus robust fitting, gate selection, and locked testing occupy strictly ordered intervals.

Table 4: Temporal partitions and permitted use.

Partition	Period	Permitted use
Training	through 31 Dec 2022	scalers; neural weights; direct XGBoost
Neural validation	1 Jan-31 Dec 2023	early stopping only
Hybrid calibration	1 Jan-30 Jun 2024	residual fit (first 70%); gate selection (last 30%)
Locked test	after 30 Jun 2024	final paired evaluation only

4 Experimental Setup

Chronological partitions. Four consecutive eras separate fitting, early stopping, residual calibration, and locked testing. Historical context may cross a boundary, but forecast targets must remain inside their assigned partition.

Comparators and training. Comparators are persistence, seasonal-naive, direct XGBoost, DLinear [11], LSTM, and TimeMixer. Static and rolling hybrid variants isolate residual safeguards. Neural models use seeds 0–4, batch size 128, AdamW at 10^{-3} , early-stopping patience seven, and at most 30 epochs.

Common-history sensitivity analysis. The primary analysis uses all pre-cutoff observations, producing unequal history lengths because BIR_A4540 begins in 2021. A pre-specified sensitivity runner restricts every station to a 2021 start while preserving all other settings; its result is not included in the present evidence.

Metrics and paired evidence. MAE is primary; RMSE, sMAPE, R^2 , bias, and fixed training-90th-percentile event metrics are secondary. A paired bootstrap resamples 24-hour blocks of

$$d_{t,h} = |y_{t+h} - \hat{y}_{t,h}^{HYB}| - |y_{t+h} - \hat{y}_{t,h}^{TM}|. \quad (12)$$

One thousand replicates estimate 95% intervals; negative differences favor the hybrid. Diebold-Mariano tests use absolute loss and Holm correction [9]. Same-origin ablations isolate embeddings, context, gating, and rolling calibration.

5 Results

5.1 The data-quality problem is material, not cosmetic

The first empirical finding is that preprocessing choices materially change the admissible forecasting sample. At MY1, the record contains 8,087 missing or non-numeric readings, 238 negative values, and a 1,656-hour maximum invalid run. Cross-station auditing reveals additional heterogeneity: HP1 contains no observations in 2016-2018; BIR_A4540 begins in 2021; CHB has only 73.44% coverage in 2024 and contains 561 negative source values; and KC1 contains an instrument-regime transition. None of these intervals or negative values is reconstructed through long-range interpolation. Instead, invalid values remain auditable and forecast windows crossing unresolved gaps are excluded.

The missingness pattern is not independent and identically distributed. Coverage changes across years and clusters by month. Consequently, complete-window evaluation samples the monitoring process as well as the pollution process. This motivates annual coverage reporting, explicit rejected-window counts, and sensitivity analysis over short-gap limits rather than a single opaque imputation pass.

5.2 The record contains structure at several forecast-relevant scales

Descriptive profiles show persistent short-lag dependence together with repeating calendar structure. Selected autocorrelations remain informative at daily and weekly lags, supporting both a 168-hour input

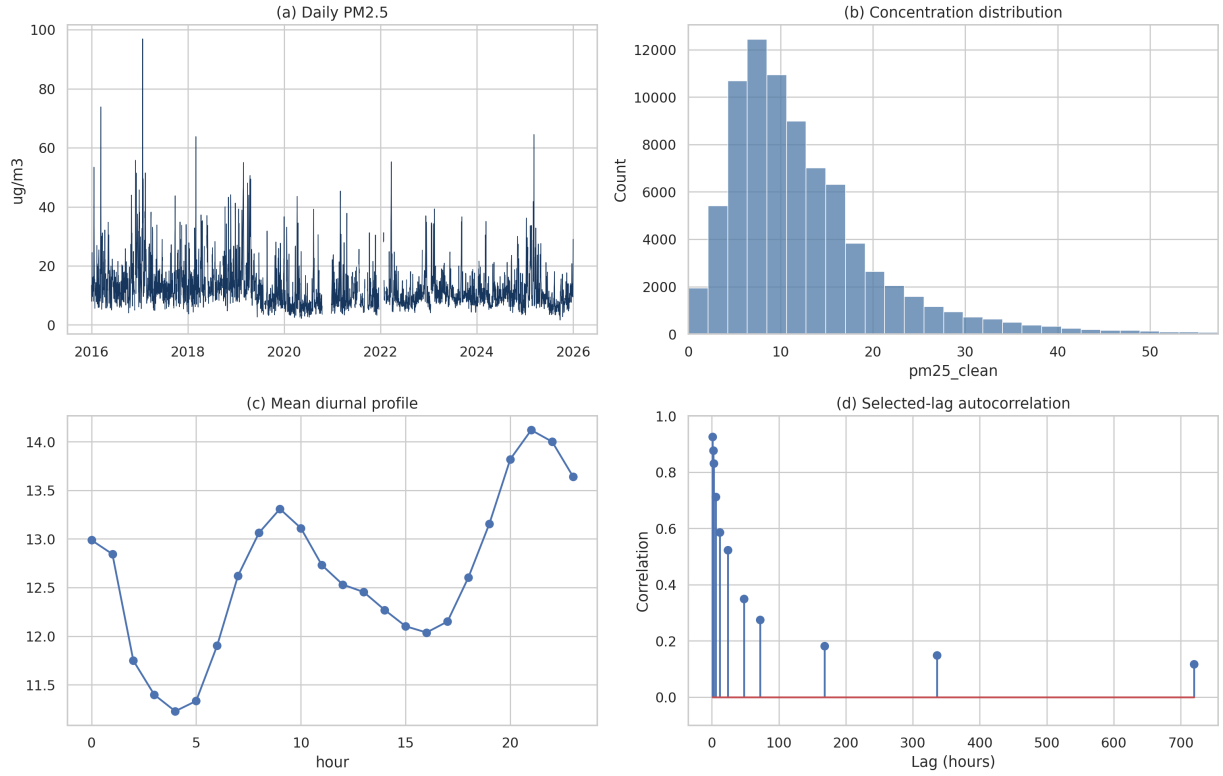


Fig. 2: EDA overview: daily PM2.5, distribution, diurnal profile, and autocorrelation.



Fig. 3: Data-quality audit: coverage, monthly missingness, gap duration, and the preserved long outage.

window and 24/168-hour engineered features. The concentration distribution is right-skewed and contains episodic peaks, so RMSE and high-event metrics complement MAE. The distribution also differs across TEOM FDMS and BAM eras. Because instrument replacement and calendar time move together, this difference cannot be attributed to the instrument alone; instrument type is therefore used only as a

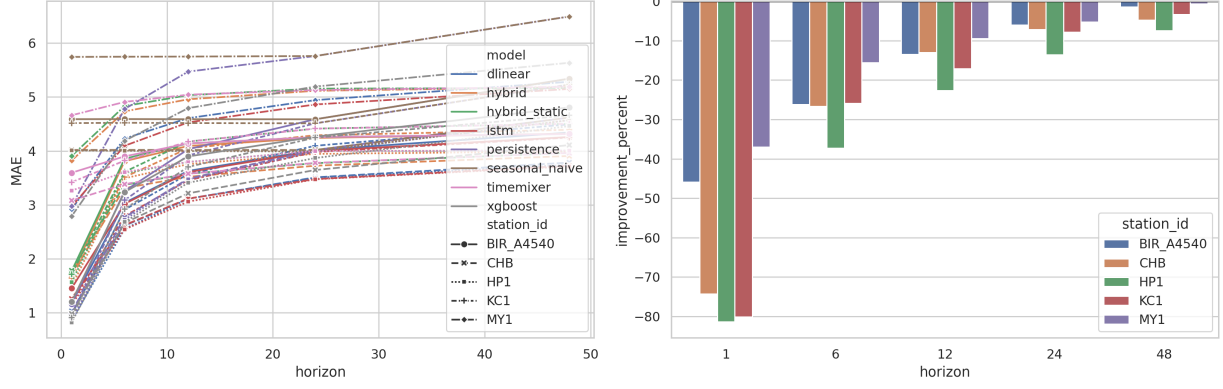


Fig. 4: Five-seed multi-station effectiveness. The hybrid is compared with the strongest non-hybrid comparator at each station and horizon.

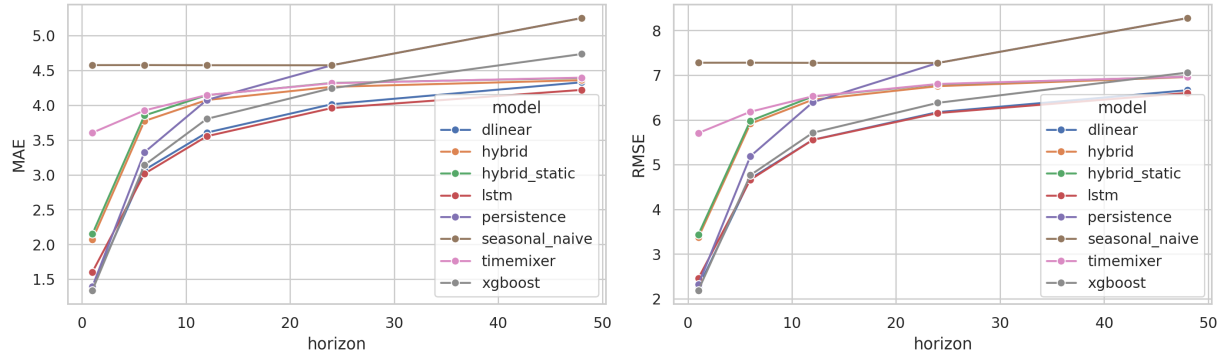


Fig. 5: Locked-test MAE and RMSE by horizon. Final claims must use the completed five-seed multi-station tables rather than a single run.

predictive regime marker.

The nine single-purpose EDA plots are retained in `figures/supplementary/` and are generated by the same plotting code.

5.3 Locked-test forecast accuracy and statistical evidence

The earlier ungated multi-station diagnostic found that rolling calibration reduced the damage caused by static residual over-correction but did not beat the strongest matched comparator. That diagnostic motivated the calibration-gated revision and is archived separately rather than reused as evidence. The result exporter accepts only complete gated-v2 runs covering all five stations, seeds 0-4, and the 30-epoch configuration. The hybrid is called effective only when it improves on the strongest matched comparator without worsening bias, high-event behavior, or seed stability.

Table 5: Locked-test MAE pooled over station-seed runs. Parentheses show dispersion across both stations and seeds; station-specific comparisons are retained in the machine-readable results.

Model	1 h	6 h	12 h	24 h	48 h
dlinear	1.390 (0.791)	3.070 (0.621)	3.607 (0.558)	4.014 (0.537)	4.329 (0.574)
hybrid	2.070 (0.902)	3.773 (0.519)	4.075 (0.502)	4.264 (0.489)	4.359 (0.472)
hybrid_static	2.149 (0.906)	3.853 (0.531)	4.144 (0.515)	4.320 (0.494)	4.391 (0.460)
lstm	1.604 (0.708)	3.016 (0.578)	3.555 (0.541)	3.959 (0.519)	4.220 (0.536)
persistence	1.394 (0.816)	3.324 (0.769)	4.076 (0.748)	4.574 (0.655)	5.251 (0.709)
seasonal_naive	4.576 (0.645)	4.577 (0.648)	4.574 (0.650)	4.574 (0.655)	5.251 (0.709)
timemixer	3.605 (0.577)	3.925 (0.540)	4.146 (0.518)	4.316 (0.483)	4.396 (0.469)
xgboost	1.341 (0.751)	3.140 (0.576)	3.804 (0.559)	4.244 (0.540)	4.735 (0.519)

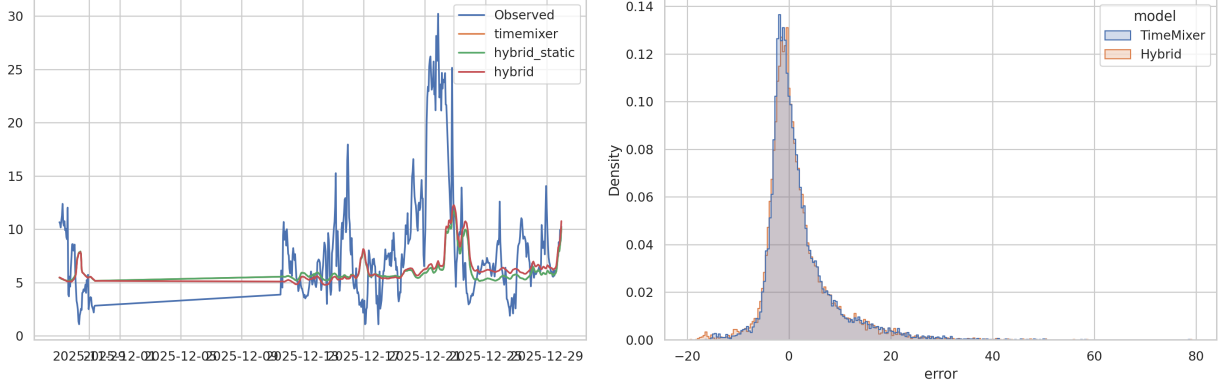


Fig. 6: Forecast-behavior diagnostics at the 24-hour horizon. The trace and residual views expose smoothing and peak-miss behavior that is not visible from a single average-error statistic.

The gated rolling hybrid won 0 of 25 matched station-horizon comparisons. Its mean relative MAE improvement against the strongest non-hybrid comparator was -23.33%.

Relative to its intended TimeMixer base, the gated rolling hybrid reduced pooled MAE at every horizon: by 42.58% at 1 hour, 3.86% at 6 hours, 1.72% at 12 hours, 1.20% at 24 hours, and 0.84% at 48 hours. Thus the residual stage was effective as a correction to TimeMixer, especially at the shortest horizon, even though it did not become the strongest model in any of the 25 station-horizon comparisons. LSTM had the lowest pooled MAE overall (3.271), while the hybrid achieved 3.708 and TimeMixer achieved 4.078.

5.4 Forecast behavior beyond average error

Average error can hide peak underprediction. The fixed training-set 90th-percentile threshold is therefore evaluated through event MAE, recall, and false-alarm rate. The 24-hour trace and residual views complement these metrics by exposing smoothing and peak-miss behavior that is not visible from a single aggregate error statistic.

5.5 What the ablations and SHAP summaries establish

The ablation study is intentionally same-origin and chronological. It distinguishes the TimeMixer base, ungated robust correction, calibration-gated static correction, rolling correction, and reduced-feature variants. This allows the paper to separate nonlinear residual fit, protection against over-correction, and adaptation to recent drift.

The ablation identifies over-correction as the main failure mode of the ungated residual learner. At horizons 12, 24, and 48 hours, its pooled MAE was 6.020, 6.420, and 6.632, respectively. Calibration gating reduced these values to 4.144, 4.320, and 4.391, and rolling calibration further reduced them to 4.075, 4.264, and 4.359. The gate selected zero correction in 56.8% of all station-seed-horizon fits. Mean gate weight was 0.930 at 1 hour but no more than 0.158 at longer horizons, showing that strong residual correction was supported primarily at short lead time.

SHAP summaries describe how the fitted residual booster allocates prediction contribution across base forecasts, embedding coordinates, and origin context. Because embedding dimensions are learned latent coordinates, interpretation is primarily grouped: total base-forecast contribution, total embedding contribution, and named context contribution by horizon. These summaries are predictive diagnostics and must not be interpreted as causal effects of instrument or calendar variables on pollution.

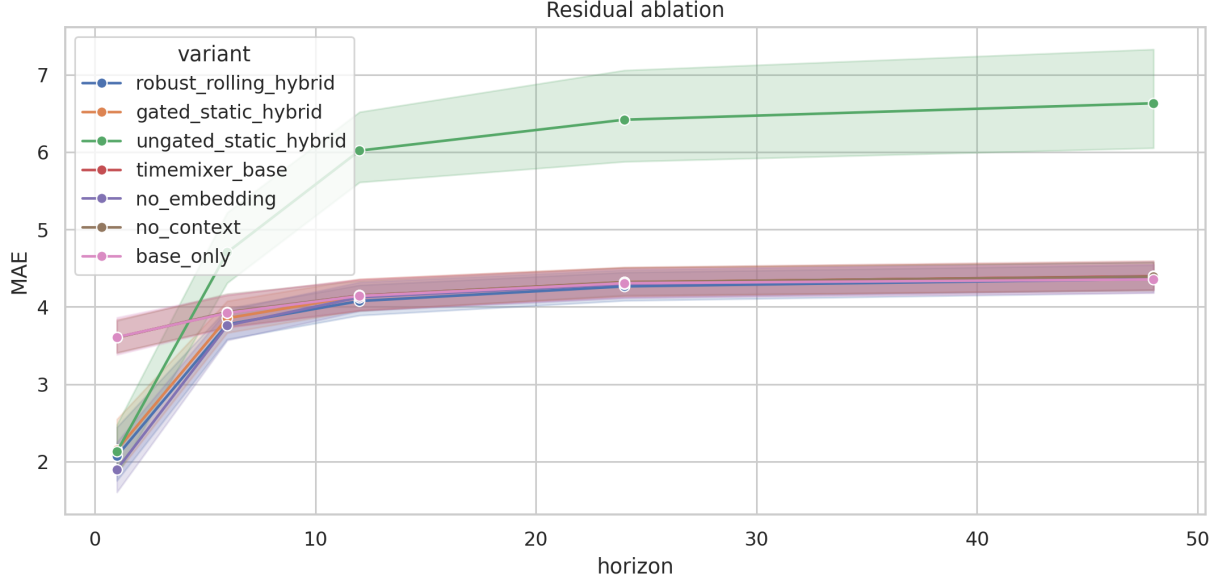


Fig. 7: Residual ablation separating the TimeMixer base, ungated robust correction, held-out calibration gating, rolling calibration, and reduced feature variants. SHAP detail is moved to the supplement.

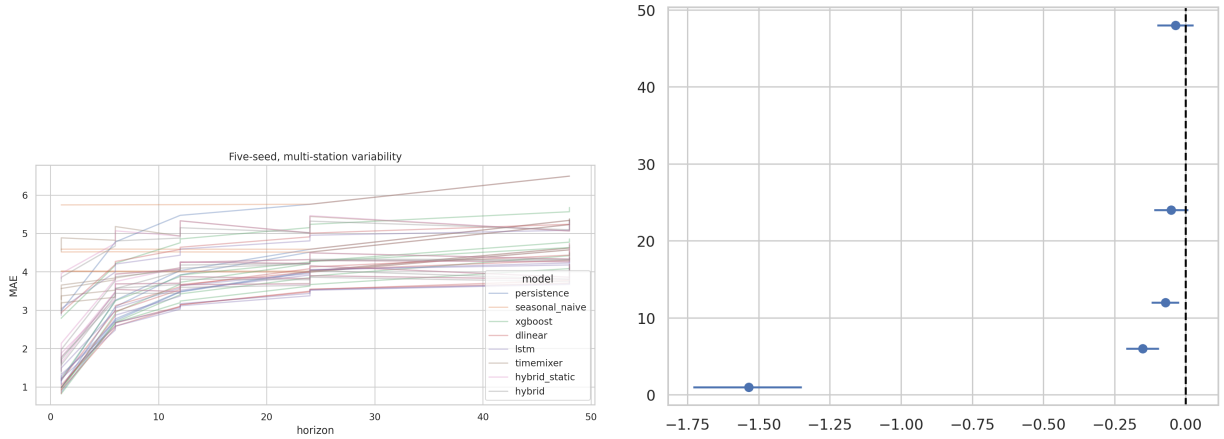


Fig. 8: Five-seed variability and paired moving-block bootstrap evidence.

5.6 Five-seed experimental setting

The confirmatory experiment uses seeds 0-4 and a maximum of 30 epochs with early stopping. Results are reported as mean and standard deviation across seeds, while station-horizon effectiveness is measured against the strongest non-hybrid comparator at the same station and horizon.

6 Discussion

Across all station-seed-horizon rows, the hybrid pooled MAE was 3.708, compared with 4.078 for TimeMixer and 3.271 for LSTM. The held-out gate selected zero correction in 56.8% of fits and had mean alpha 0.24, quantifying how often fallback protection was needed. Because gate selection did not access the locked test, the remaining gap reflects generalization rather than post-hoc test selection. The method therefore achieved its narrow goal of improving TimeMixer while failing the stricter goal of beating the best model available at each station and horizon.

Effectiveness is therefore narrow rather than global: 81 of 125 paired bootstrap intervals favor the hybrid over TimeMixer, but none of the 25 station-horizon comparisons beats its strongest alternative. At one hour, recent lags support a large correction; at longer horizons the gate usually shrinks toward zero. LSTM

and DLinear then benefit from stronger base trajectories, suggesting that exogenous forecasts or a stronger backbone may matter more than additional booster capacity.

7 Limitations and Threats to Validity

Inference and construct validity. Overlapping windows retain serial dependence despite chronological splitting and block inference. The locked test must not guide later tuning. Station-level PM2.5 is not personal exposure, and the training-set 90th percentile is an analytical event threshold rather than a health category.

External validity and data selection. Five UK stations improve on a single-site study but do not establish transfer to unseen, industrial, or non-UK sites. Unequal histories and outages alter the admissible sample: HP1 lacks 2016–2018, BIR_A4540 begins in 2021, and CHB has reduced 2024 coverage. The common-history sensitivity runner is provided but its result is not part of the present evidence.

Measurement and omitted drivers. Instrument indicators may confound sensor response with calendar changes and are not causal. Historical PM2.5 alone also omits meteorology, traffic, co-pollutants, transport, and event information, limiting abrupt-episode prediction.

Use boundary. The model is not an official warning system. Deployment should disclose missingness, uncertainty, instrument status, update time, and abstention conditions rather than replace monitoring or public-health guidance.

8 Future Work

Future evaluation should combine leave-one-station-out transfer with industrial and suburban sites, add forecast-available meteorology and regional transport variables, and test change-point or regime-specific calibration. All extensions must retain origin-safe inputs and locked testing.

9 Conclusion

The leakage-controlled workflow preserves long gaps and instrument regimes while separating neural training, residual calibration, and locked testing.

The gated hybrid improved its TimeMixer base at all five horizons and reduced the severe over-correction of the ungated residual model. However, it won 0 of 25 comparisons against the strongest matched non-hybrid comparator and had mean relative improvement of -23.33% under that stricter reference. LSTM was the best pooled model (MAE 3.271). The evidence supports calibration gating as a useful safeguard, but it does not support overall superiority or a state-of-the-art claim.

Data and Code Availability

The repository contains data-processing and modeling code, tests, figures, experiment outputs, and manuscript source. UK-Air attribution and licensing conditions must be preserved.

Conflict of Interest

The author declares no conflict of interest.

References

- [1] World Health Organization. WHO global air quality guidelines: Particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. World Health Organization, 2021.
- [2] Shiyu Wang, Haixu Wu, Xiaoming Shi, Tengge Hu, Huakun Luo, Lintao Ma, James Y. Zhang, and Jun Zhou. Timemixer: Decomposable multiscale mixing for time series forecasting. In *The Twelfth International Conference on Learning Representations*, 2024.
- [3] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- [4] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.
- [5] Yun Bai, Bo Zeng, Chuan Li, and Jin Zhang. An ensemble long short-term memory neural network for hourly pm_{2.5} concentration forecasting. *Chemosphere*, 222:286–294, 2019.
- [6] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations*, 2023.
- [7] Jie Yang, Rui Yan, Mingyue Nong, Jiaqiang Liao, Feipeng Li, and Wei Sun. Pm_{2.5} concentrations forecasting in beijing through deep learning with different inputs, model structures and forecast time. *Atmospheric Pollution Research*, 12(9):101168, 2021.
- [8] Cort J. Willmott and Kenji Matsuura. Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance. *Climate Research*, 30(1):79–82, 2005.
- [9] Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995.
- [10] Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.
- [11] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 11121–11128, 2023.